Data rate = Sample rate x # of bits/sample

$N = f_s \times n_b$

PCM BW,

Let for NRZ,

$r = 1, c = \frac{1}{2}$

$BW_{min} = c \times N \times \frac{1}{r}$

$= \frac{1}{2} \times f_s \times n_b \times \frac{1}{1}$

$= \frac{f_s \times n_b}{2}$

upto this mid...

Ch-5_1

27-07-26

Digital-to Analog

Here analog conversion, bandpass channel needed ($BW \neq 0$)

Digital data ⟶ Analog signal

by Carrier frequency (f_c)

$s(t) = A \sin(2\pi f t + \phi)$

Amp (ASK) — Freq (FSK) — Phase (PSK) — conversion techniques (not used now)

QAM (Quadrature amplitude modulation)

Implementation process, BW Analysis

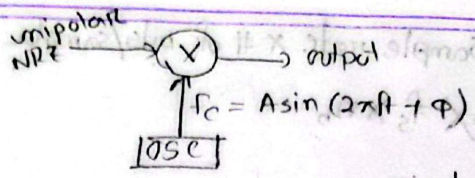
for this case,

$r = \frac{\#\text{ data element}}{\#\text{ signal element}} = \#\text{ data element}$ (always 1)

& $S = N \times \frac{1}{r}$ bauds

for $n = 1$, $S = N$

$n = 2, 3, \ldots$, $S < N$



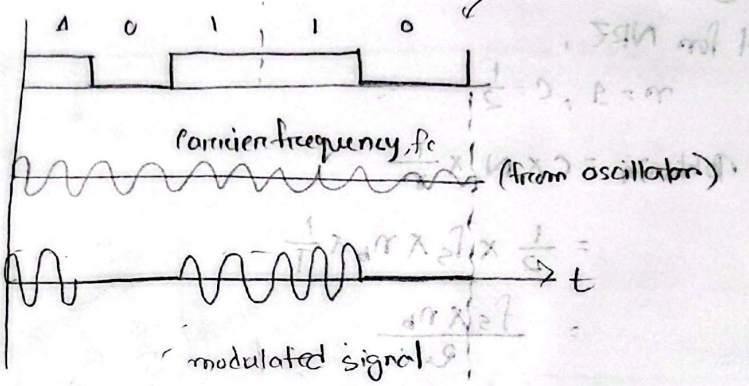
ASK (BASK)

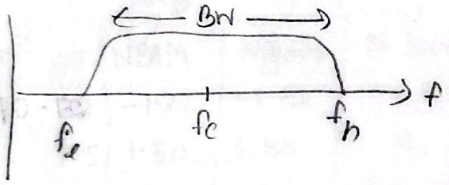
$n_b = 1$

$2^1 = 2 \begin{cases} 0 \\ 1 \end{cases}$

Amp vary করবে, (0/1)

phase & freq. const. থাকবে।





→ It is a non periodic composite signal. So, in freq domain, we get continuous.

$f_1 = f_c - S/2 = f_c - N/2$

$f_h = f_c + S/2 = f_c + N/2$

for $n = 1$ (signal element always 1)

$S = N \times \frac{1}{n}$ (we don't use case factor here)

$= N$

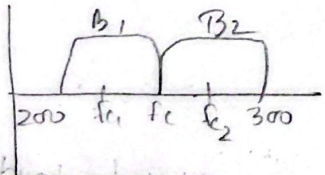
$BW \propto S$

$BW = (1+d)S$

↳ Modulation factor

$d = 0$; $BW_{min} = (1+0)S = S = N$

$d = 1$; $BW_{max} = (1+1)S = 2S = 2N$



B_1, $N = ?$ B_2, $N = ?$

1/2 হয়ে গেল, full duplex করলে 2 টা

29-07-26

FSK (Frequency Shift keying)

- (Binary FSK)
  - Non Coherent
    - ↳ Discontinuity in phase
  - Coherent
    - ↳ Continuity in phase
- (Multilevel FSK) — MFSK

$n_b = 2, 3, \ldots$

$2^2 = 4$

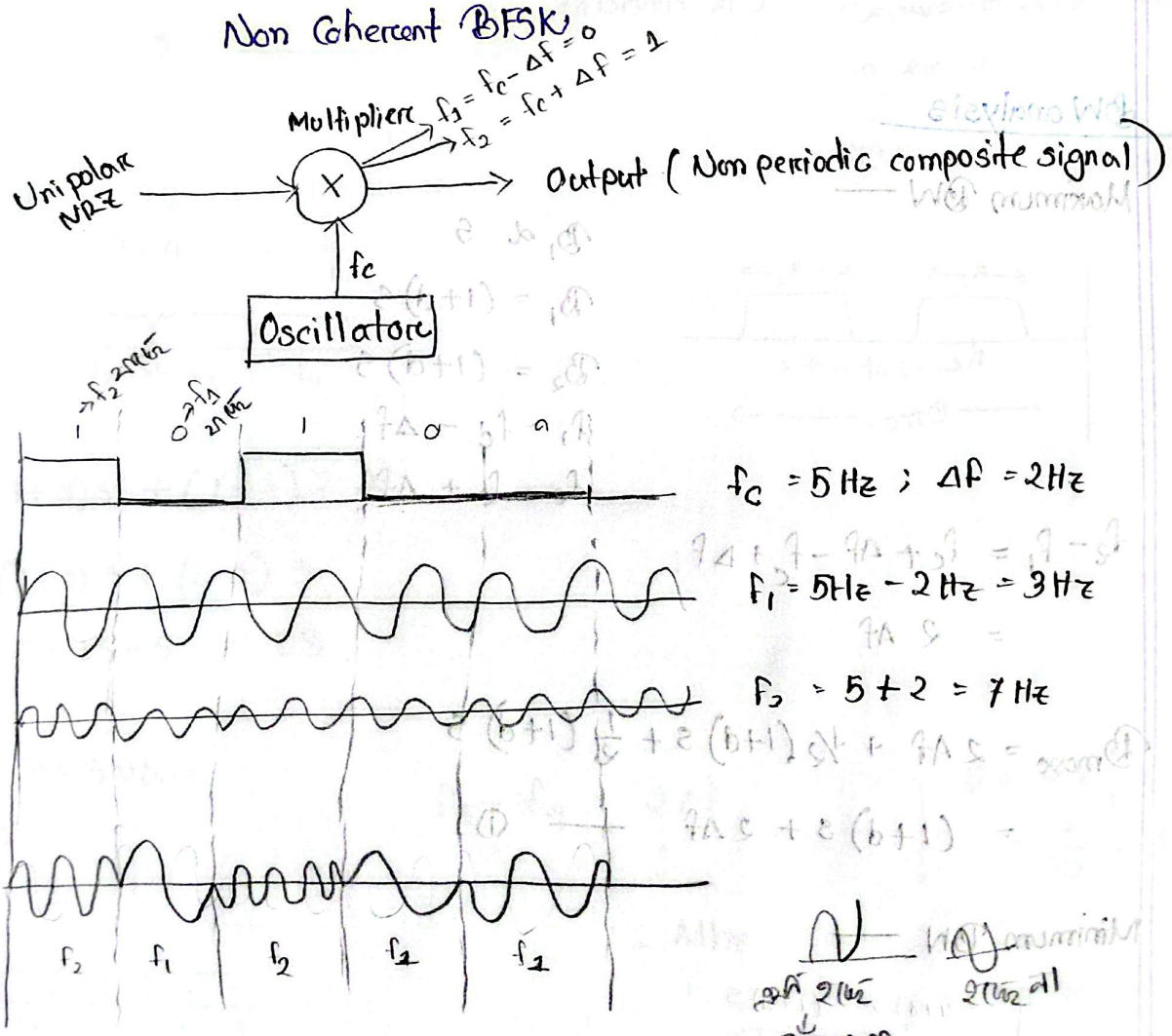
$00 \rightarrow f_1$

$01 \rightarrow f_2$

$10 \rightarrow f_3$

$11 \rightarrow f_4$

Non Coherent BFSK

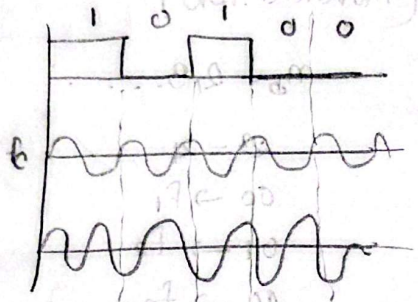


2Hz এর 2Hz এর non coherent

coherent

unipolar → VCO → output

↳ VCO দিয়ে low & high freq generate করা ।

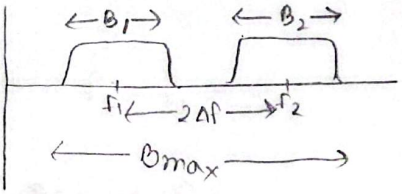


↳ phase discontinuity নাই ।

এই coherent ।

BW analysis

Maximum BW —



$B_1 \propto S$

$B_1 = (1+d)S$

$B_2 = (1+d)S$

$f_1 = f_c - \Delta f$

$f_2 = f_c + \Delta f$

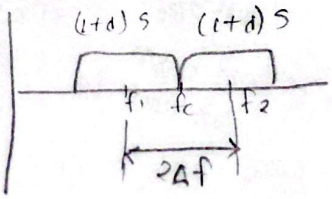
$f_2 - f_1 = f_c + \Delta f - f_c + \Delta f$

$= 2\Delta f$

$B_{max} = 2\Delta f + \frac{1}{2}(1+d)S + \frac{1}{2}(1+d)S$

$= (1+d)S + 2\Delta f$ —— (i)

Minimum BW —



$2\Delta f = \frac{1}{2}(1+d)s + \frac{1}{2}(1+d)s$

$= (1+d)s$ [for minimum BW, $d=0$; $r=1$; $S=N$]

$= (1+0)N$

$= S = N$

From eqn – (1)

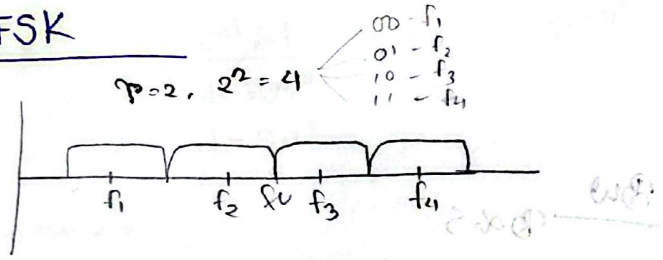
$B_{min} = (1+0)s + 2\Delta f$

$= s + s$

$= 2s$ ——————— (2)

Derivation → can come in exam.

MFSK



$B_{max} = (1+d)s + (L-1)2\Delta f$

$B_{min} = (1+0)s + (L-1)s$

$= s + Ls - s = Ls$

~~x~~ $r = 3$; $N = 3$ Mbps

$L = 2^r = 8$

$f_5 - f_4 = 2\Delta f$

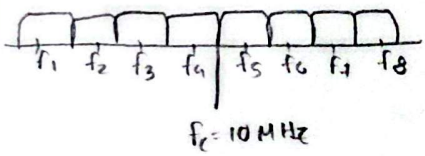
$= S$

$= 1$ MHz

↳ $2\Delta f$ A assign করা যায় the 2টা বোঝে।

$S = N \times \frac{1}{r}$

$= \frac{3}{3} = 1$ Mbaud



$\therefore f_4 = 9.5$ MHz

$f_5 = 10.5$ MHz

$B_{min} = Ls = 8 \times 1$ Mbaud

$= 8$ MHz

30-07-26

PSK (Phase shift keying)

BPSK (Binary PSK)

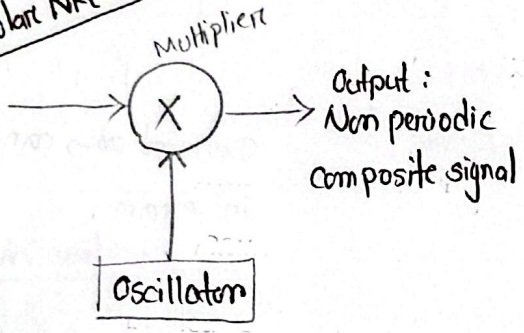
$n_b = 1 ; 2^1 = 2$ < 1 – 0°, 0 – 180°

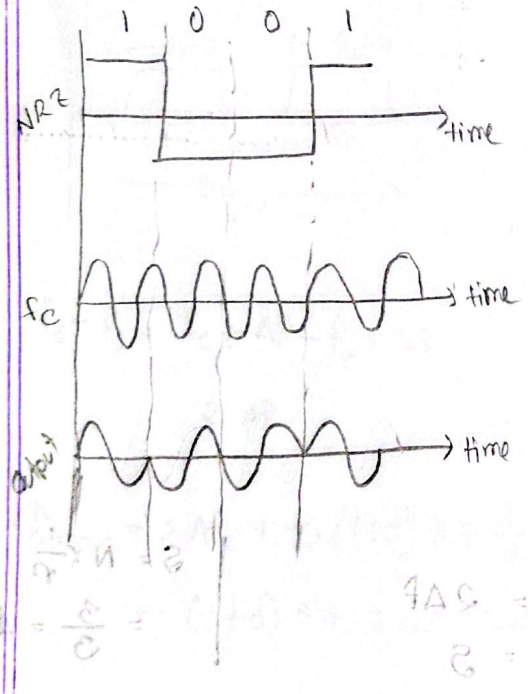
QPSK (Quadrature PSK)

$n_b = 2, 3, \ldots$

for $n_b = 2 ; 2^2 = 4$

Polar NRZ-L

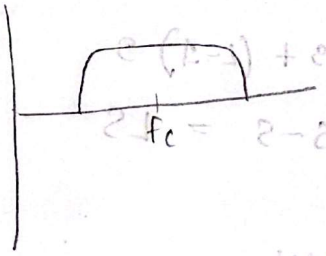




Bw

$B \propto S$

$B = (1+d)S$

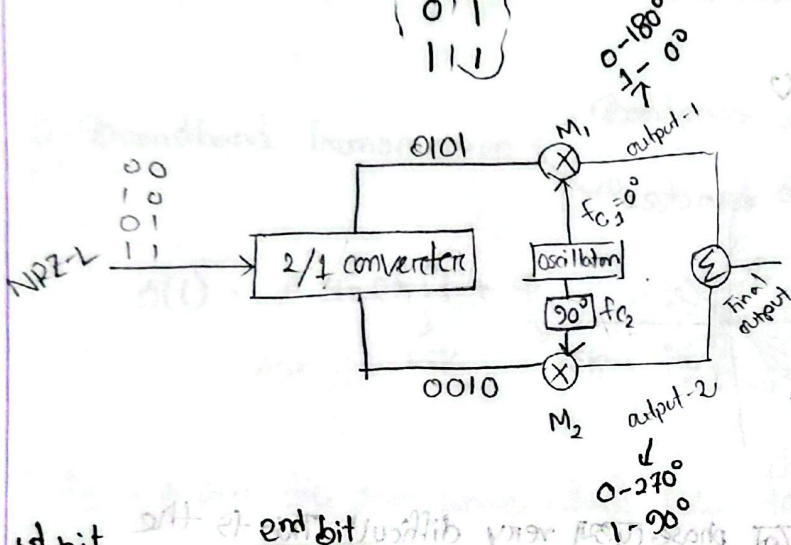


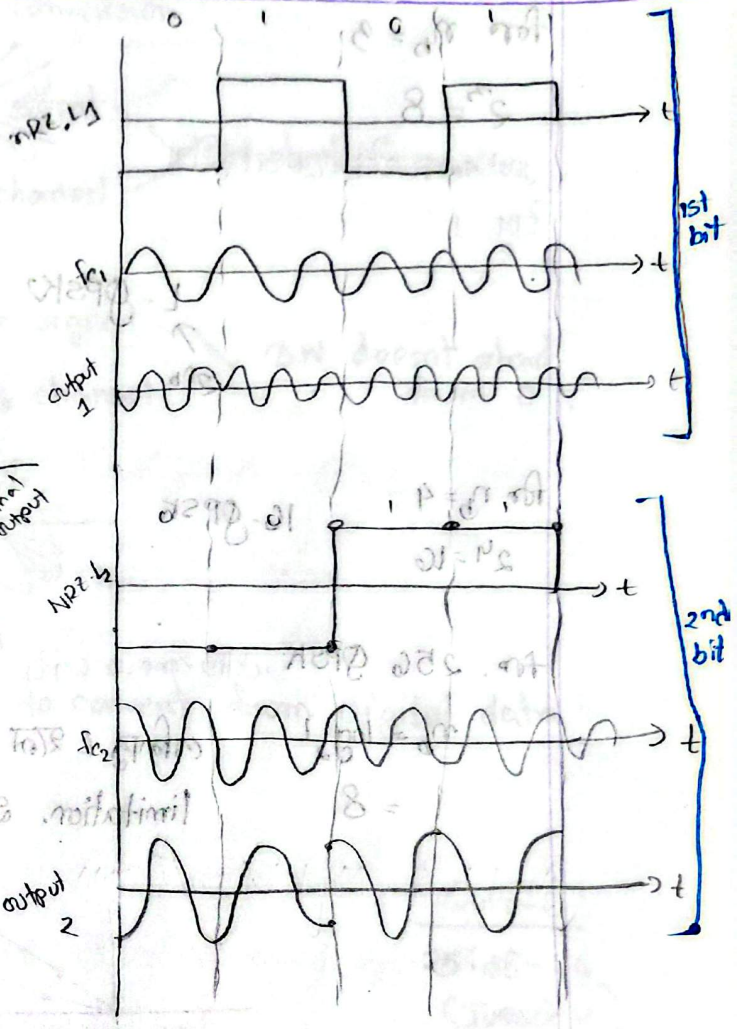
QPSK

$n_b = 2, 3, \ldots$

for $n_b = 2$, $Q^2 = 4$,

1st bit	2nd bit
0	0
1	0
0	1
1	1





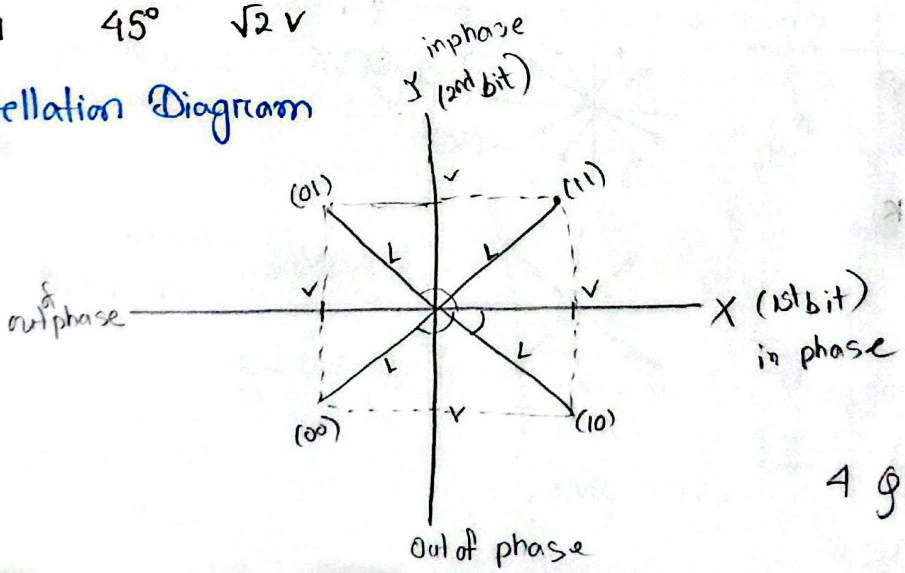
1st bit	2nd bit
0 - 180°	0 - 270°
1 - 0°	1 - 90°

After combining

	Phase	Amp
0 0	-135°	√2 V
1 0	-45°	√2 V
0 1	135°	√2 V
1 1	45°	√2 V

→ 4-QPSK

Constellation Diagram



$L^2 = V^2 + V^2$

$L = \sqrt{2V^2}$

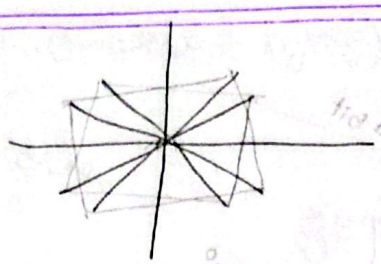
4 QPSK

for $n_b = 3$

$2^3 = 8$

just কলম 2টা করে bit প্রতি

দেয়া।



⇒ 8-QPSK

L-QPSK

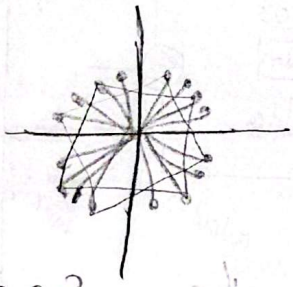
↗

2^{n_b}

for $n_b = 4$

$2^4 = 16$

16-QPSK



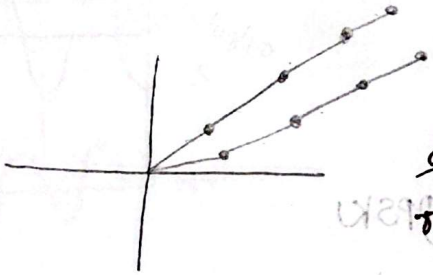
for. 256 QPSK, প্রতি zone এ 64টা

$n_b = \log_2 L$

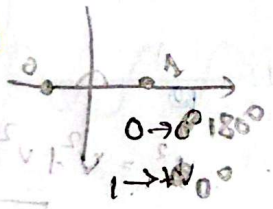
$= 8$

কিন্তু L হলে phase দেওয়া very difficult. This is the limitation. So, we do, PSK + ASK

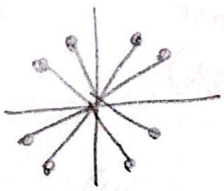
QAM



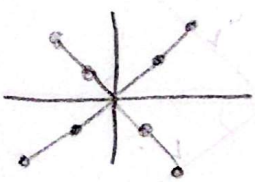
একই phase এ diff amp. দিয়ে represent করা।



ASK, PSK, QPSK



8-QPSK অনেকগুলা phase Amp same.



8-QAM Amp vary same phase

17-08-26

Digital to Analog Conversion

1. Baseband Transmission < baseband signal / low pass channel > BW starts from 0

2. Broadband Transmission < Bandpass signal / Passband channel > BW doesn't start from 0.

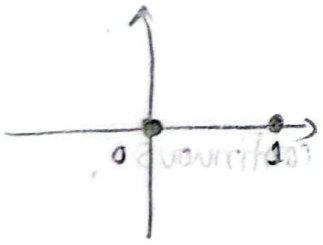
$s(t) = A \sin 2\pi ft + \phi$

ASK — FSK — PSK

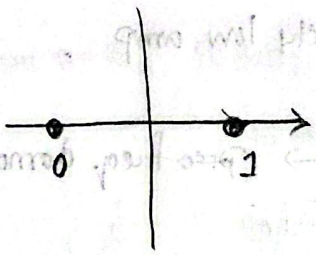
f_c = a periodic sign wave which helps to convert from digital data to analog signal.

$f_c = A \sin 2\pi ft + \phi$

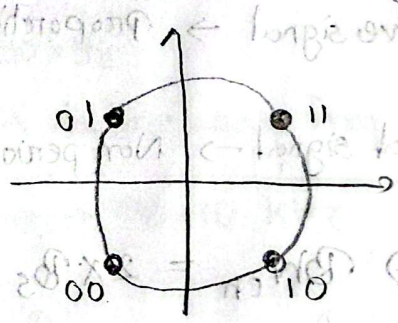
Quiz-2
25-08-26
(Tuesday)



ASK

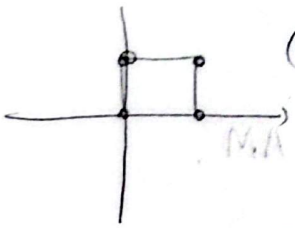


BPSK



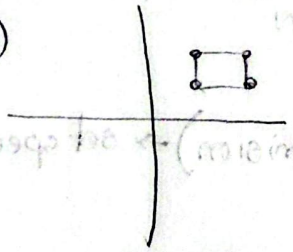
QPSK

2-QPSK
4-QPSK
4 QAM

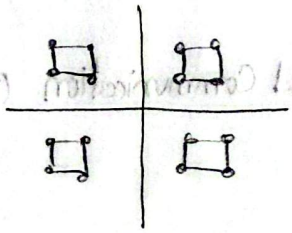


(QPSK + ASK)

4 QAM but Not 4 QPSK



4 QAM

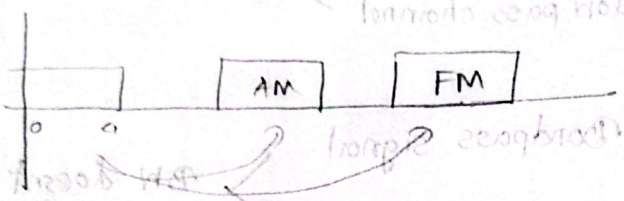


16 QAM

ch 5.2 19-08-26

Analog Transmission

Analog to Analog :



Analog to Analog transmission is done to send in a specific BW.

$f_c = S(t) = A \sin(2\pi f t + \Phi)$

- AMP (AM) → Found
- Freq. (FM) → Found
- phase (PM) → Not found

Amplitude Modulation (AM)

Positive signal → proportionately high amp

Negative signal → proportionately low amp .

Final signal → Non periodic → ~~pre~~ freq. domain → continuous .

$AM \rightarrow BW_{ch} = 2 \times B_s$

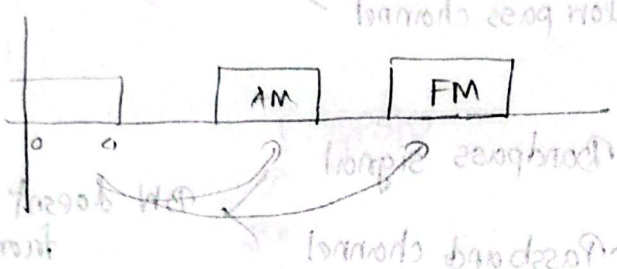
- BW_{ch} : channel bw
- B_s : signal sent at bw

FCC (Federal Communication Commision) → set spectrum of AM .

ch 5.2 19-08-26

Analog Transmission

Analog to Analog :



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$$AM \rightarrow BW_{ch} = 2 \times B_s$$

- BW_{ch} : channel bw
- B_s : signal sent आई BW

FCC (Federal Communication Commision) → set spectrum of AM.

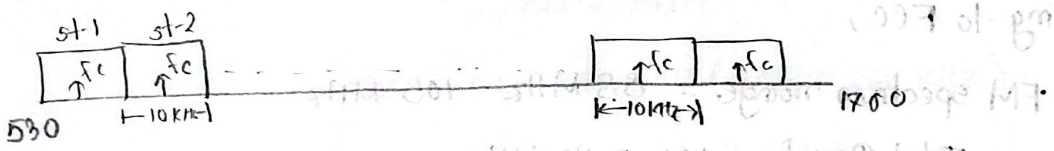
According to FCC, AM spectrum range = 530 kHz – 1700 kHz

Total BW for AM = 1700 kHz − 530 kHz = 1170 kHz

FCC, ad. Audio Bs = 5 kHz

channel BW for AM, $BW_{ch} = 2 \times Bs = 10$ kHz for each station.

$$\#\text{ of AM station} = \frac{\text{Total BW}}{\text{Channel BW}} = \frac{1170\ \text{kHz}}{10\ \text{kHz}} = 117 \text{ stations}$$



st-1 : Spectrum range : 530 kHz to 540 kHz

: carrier freq : $530\ \text{kHz} + \frac{10}{2}\ \text{kHz} = 535$ kHz

st-n : Spectrum range : $(n-1) \times 10\ \text{kHz} + 530\ \text{kHz}$

for $n = 2$, $10\ \text{kHz} + 530\ \text{kHz}$

$= 540$ kHz → starting range freq

∴ ending freq → 540 + 10 kHz

$= 550$ kHz

∴ Range → 540 ~ 550 kHz

Frequency Modulation :

+ve signal → high freq

-ve signal → low freq

$B_{FM} = 2(1+\beta) B_s$

$BW_{cn} = 2(1+\beta) B_s$

$\beta = 4$

According to FCC,

FM spectrum range = 88 MHz – 108 MHz

Total BW for FM = 20 MHz

FCC, Audio B_s = 15 kHz

channel BW for FM, $BW_{cn} = 2(1+\beta) B_s = 2 \times 5 \times 15 + (\quad)$

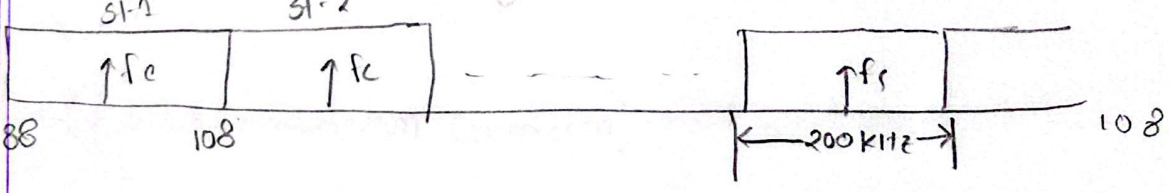
= 150 kHz + Guard band (50 kHz)

↳ unused frequency where nothing passed so, overlap doesnot occur

= 200 kHz

$$\# \text{ of FM station} = \frac{\text{Total BW}}{\text{channel BW}} = \frac{20 \text{ MHz}}{200 \text{ kHz}}$$

= 100 stations



st-1 : spectrum range = 88 MHz ~ (88 + 200 KHz)

= 88 ~ 88.2 MHz

carrier freq : f_{c_1} = 88 MHz + 100KHz

= 88.1 MHz

st-n spectrum range = (n-1) x 200KHz + 88 MHz

for st_2 → = 200 KHz + 88 MHz

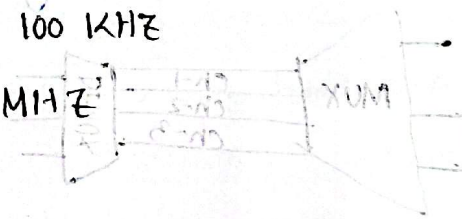
= 88.2 MHz

range: 88.2 MHz ~ (88.2 + 200 KHz)

= 88.2 ~ 88.4 MHz

f_{c_2} = 88.2 + 100 KHz

= 88.3 MHz



For PM → B = 2

24-08-26

Ch-6 → Pairoutram

Multiplexing

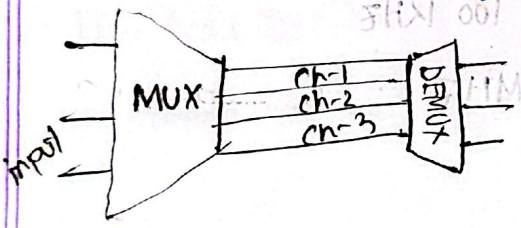
Allows simultaneous transmission of multiple signals in a single data link.

↳ Physical path

Multiplexing

- Frequency division multiplexing (analog)
- Time division multiplexing (digital)

Frequency division Multiplexing



link ← ch1, ch2, ch3 — Frequency divided

To input to MUX, check

↳ 1. Input lines are analog / digital

2. If analog, then ok, go to 5

3. If digital then convert to analog (ASK, FSK, PSK, QPSK, QAM)

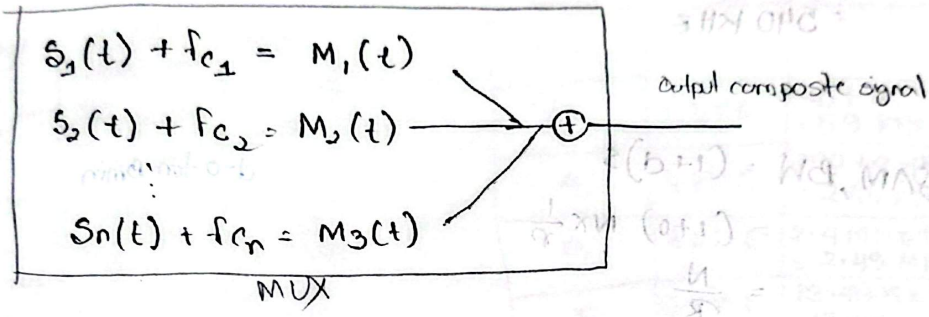
4. Find input line BW

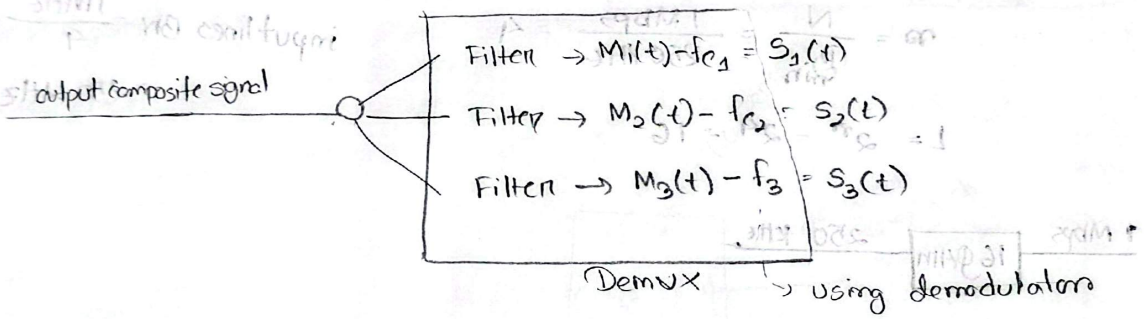
5. All input lines BW ≤ link BW

BW → diff. value ✓

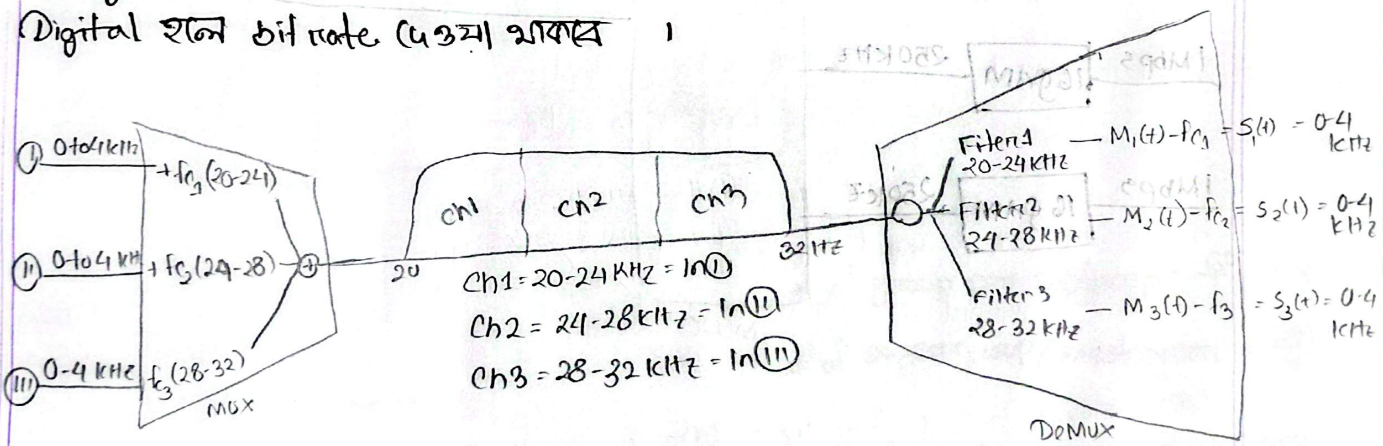
Spectrum → high and low → range ✓

signal গুলোর spectrum range এর সমান channel spectrum range assign করতে হবে । by FM modulation





Analog হলে BW দেওয়া থাকবে ।
Digital হলে bit rate দেওয়া থাকবে ।



These needs guard band. But here we cannot. So, to add guard band 1 KHz, we need (n-1) guard bands. So, Link BW needs to be 12 + (3-1)1 = 14 KHz

Eg- 6.2 $5 \times 100\ KHz \rightarrow (5-1)10\ KHz$

$= 500 + 40\ KHz$

$= 540\ KHz$

Eg- 6.3

QAM, BW $= (1+d)S$

$= (1+0)\ N \times \frac{1}{n}$

$= \frac{N}{R}$

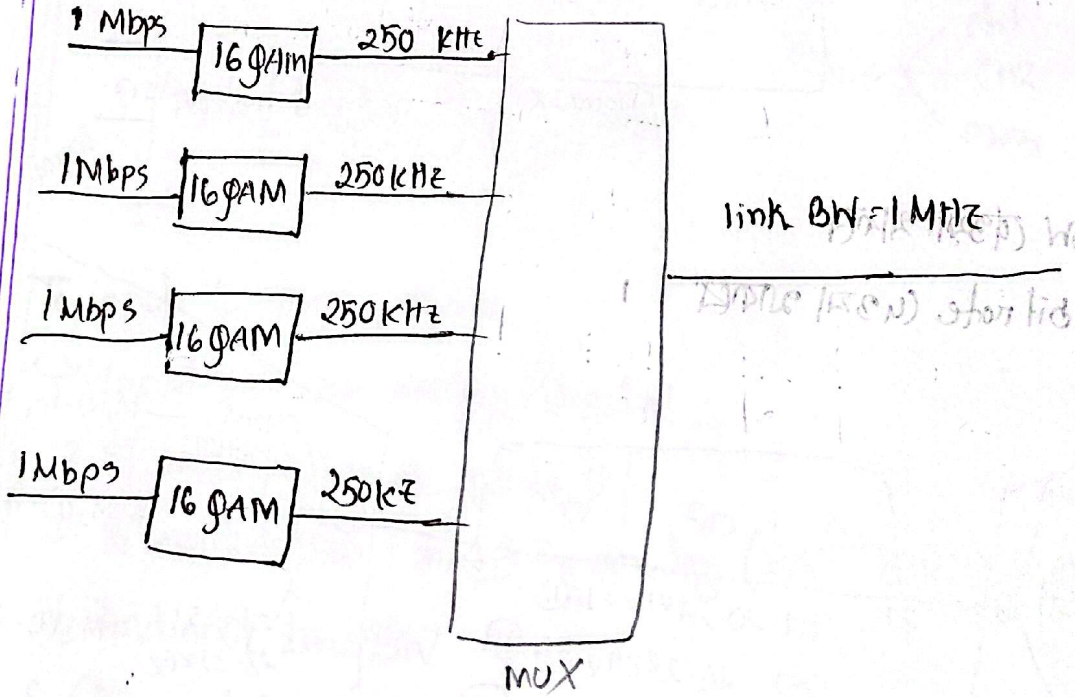
$d = 0$ for B_{min}

$n = \frac{N}{B_{min}} = \frac{1\ Mbps}{250\ KHz} = 4$

input lines BW $= \frac{1MHz}{4}$

$= 250\ KHz$

$L = 2^n = 2^4 = 16$



27-08-26

FDM Analog Hierarchy

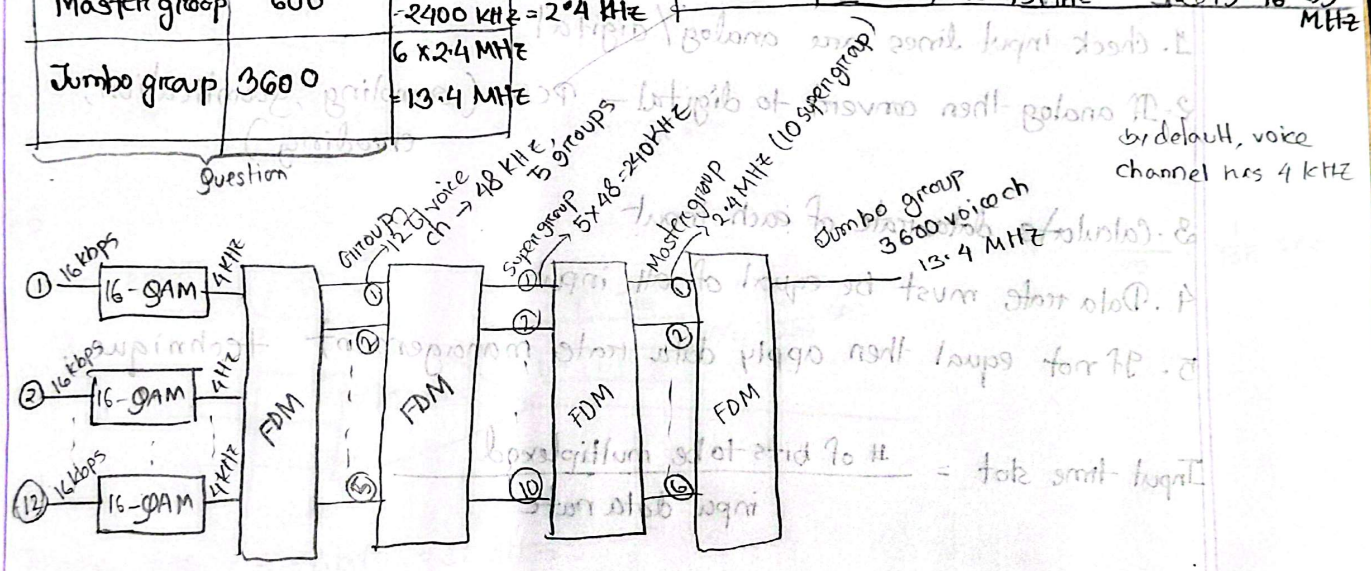
starts from 50 – given; from 1 kHz – given

Hierarchy level	# of channel	BW
Channel	1 data 16 kbps	4 kHz
Group	12	12×4 kHz = 48 kHz
Super group	60	5×48 kHz = 240 kHz
Master group	600	10×240 kHz = 2400 kHz = 2.4 MHz
Jumbo group	3600	6×2.4 MHz = 13.4 MHz

Question

# of guard band	BW for each guard band	Total BW	Spectrum range
11	11 kHz	48+11 = 59 kHz	50 ~ (50+59) = 50 ~ 109 kHz
4	4 kHz	240+4 kHz = 244 kHz	109 ~ (109+244) = 109 ~ 353 kHz
9	9 kHz	2.4 MHz + 9 kHz = 2.49 MHz	353 ~ (353+2490) = 353 kHz – 2843 kHz
5	51 kHz	13.4 + 5 kHz = 13.45 MHz	2843 kHz – 16.293 MHz = 2.843 – 16.293 MHz

by default, voice channel has 4 kHz



QAM BW = (1+d)S

= (1+0) N × 1/r [S = N × 1/r]

4 kHz = 16 kbps × 1/r

r = 16 kbps / 4 kHz = 4 bits / signal element

L = 16

# of group per super group = 60/12 = 5

# of super group / Master group = 600/60 = 10

# of Master group / Jumbo group = 3600/600 = 6

Find Total BW with guard band = 13.43 MHz

Spectrum range = 50 KHz ~ 16.29 MHz

31-08-26

TDM

time is divided.
not BW.

1. Check input lines are analog / digital
2. If analog then convert to digital – PCM (sampling, Quantization, encoding)
3. Calculate data rate of each input
4. Data rate must be equal of all input
5. If not equal then apply data rate management technique.

$$\text{Input time slot} = \frac{\text{\# of bits to be multiplexed}}{\text{inpu data rate}}$$

Frame duration = input time slot = T

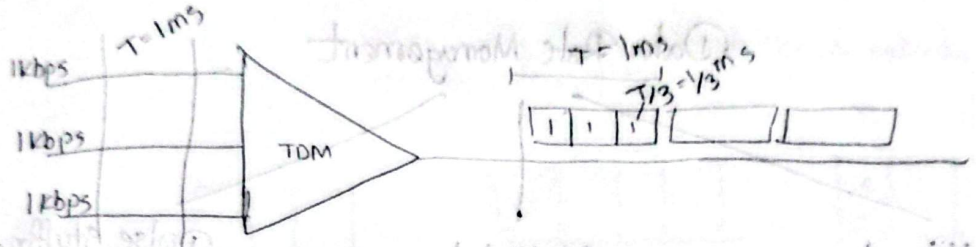
$$\text{Output time slot} = \frac{\text{Input time slot}}{\text{\# of input}} = \frac{T}{\text{\# of input}}$$

Frame size = # of output slot x # of bits / output slot

Output / link data rate = # of input x data rate / input

Frame rate = $\frac{\text{output data rate}}{\text{frame size}}$ frame/sec

eg- 6.5



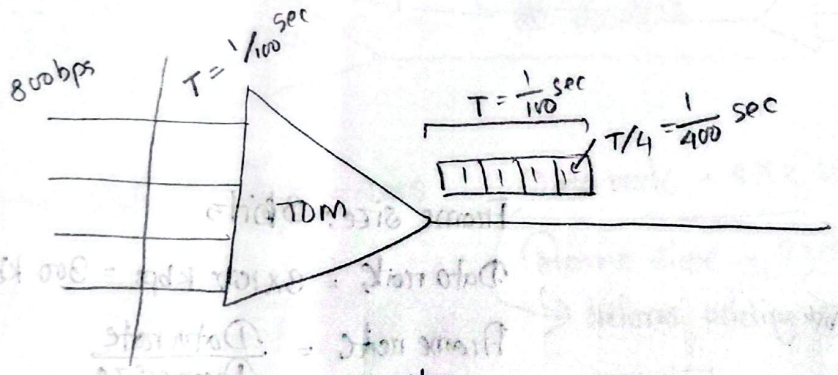
$T = \frac{1}{10^3} = 1ms$

Frame size = 3×1 = 3 bits

Data rate = 3×1 kbps = 3 kbps

Frame rate = $\frac{3\text{ kbps}}{3\text{ bits}}$ = 1 k frames/sec

6.8



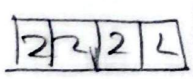
$T = \frac{8}{800} = \frac{1}{100}$ sec

Frame size = 4×8 bits = 32 bits

Data rate = 4×800 bps = 3200 bps

Frame rate = $\frac{3200\text{ bps}}{32\text{ bits}}$ = 100 bits/sec

6.9 →

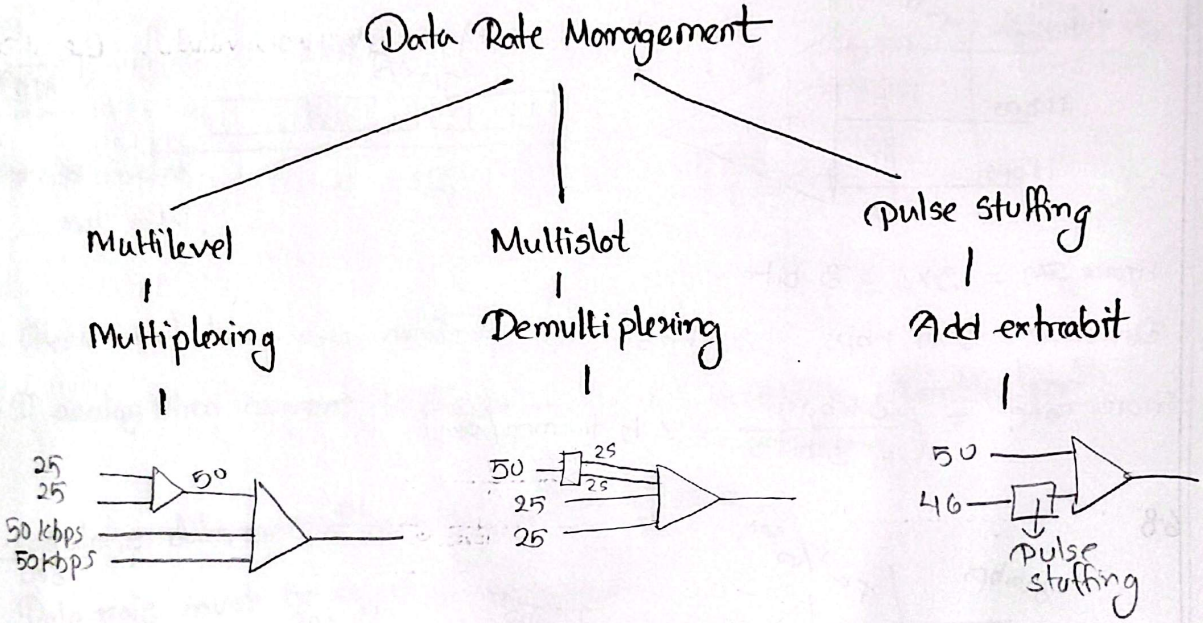


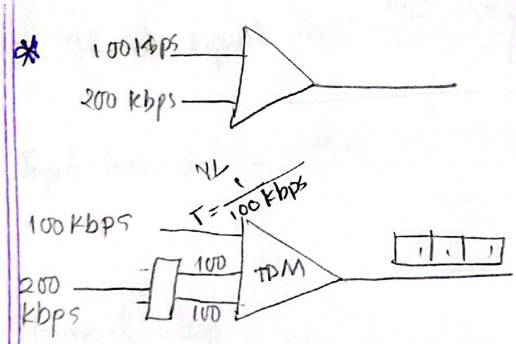
02-09-26

২টা same data rate এর হয়,

Multilevel Multiplexing

Data Rate Management





Frame size: 3 bits

Data rate = 3 × 100 kbps = 300 kbps

Frame rate = $\frac{\text{Data rate}}{\text{Frame size}}$

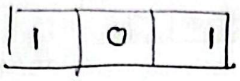
= $\frac{300 \text{ kbps}}{3 \text{ bits}}$

= 100 k frames/sec

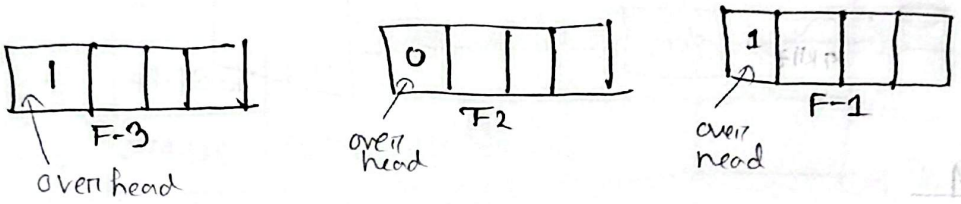
Input time slot / Frame duration = $\frac{1}{100\text{k}}$ sec

Interleaving – Bit values are placed on frames and from frames to receivers.

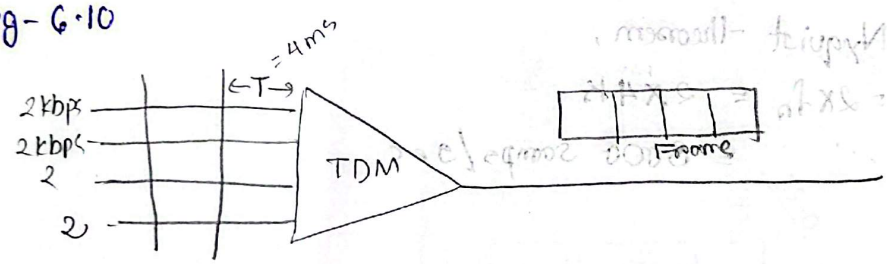
Frame Synchronization —



→ if not these values, there is error



eg- 6.10



Input time slot = $\frac{8}{2\text{kbps}}$ = 4ms

Data rate = 4×2 kbps = 8 kbps

Frame size = 4×8 = 32 bits
↳ before adding overhead.

Frame rate = $\frac{\text{Data rate}}{\text{Frame size}}$

= $\frac{8\text{ kbps}}{32\text{ bits}}$ = 250 frame/sec

# of overhead bits = 250 × 1 bit = 250 bit

After adding overhead bits, Frame size = 32 bits + 1 bits = 33 bits

Data rate = frame rate × frame size

= 250 × 33 = 8250 bps

Prev. data rate = 8000 bps

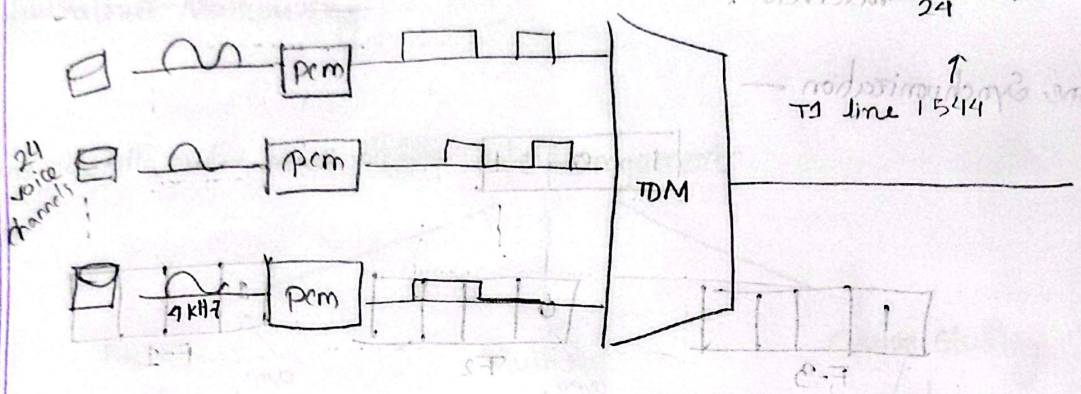
New data rate = 8250 bps

∴ Overhead rate = (8250 - 8000)

= 250 bps

bits 50 overhead 8k

$\frac{1544}{24} = 64 \cdot 33$ kbps



PCM

1) Sampling:

According to Nyquist theorem,

$f_s = 2 \times f_h = 2 \times 4K$

$= 8000$ samps/sec

2) Quantization:

# of level and bit

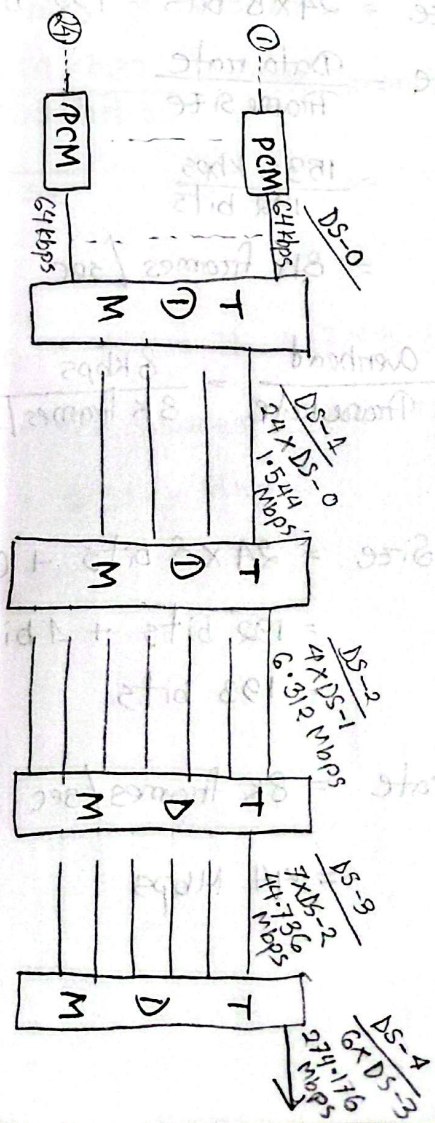
$N = f_s \times n_b$

$n_b = \frac{N}{f_s} = \frac{64 \text{ Kbps}}{8000} = 8$ bits/sample

03-09-26

Digital hierarchy

Service	Line	Data rate (Mbps)	Voice channels	Group of Service	Overhead	Overhead bits
DS-1	T-1	1.544	24	24 x DS-0	8 kbps	1
DS-2	T-2	6.312	96	4 x DS-1	168 kbps	21
DS-3	T-3	44.736	672	7 x DS-2		
DS-4	T-4	274.176	4032	6 x DS-3		



DS-1

Data rate = 24 x DS-0

= 24 x 64 kbps

= 1536 kbps

According to Table,

DS-1 Data rate = 1544 kbps

Overhead = (1544 - 1536) kbps

= 8 kbps.

Without overhead,

Frame size = 24 x 8 bits = 192 bits

Frame rate = $\frac{\text{Data rate}}{\text{Frame size}}$

= $\frac{1536 \text{ kbps}}{192 \text{ bits}}$

= 8K frames/sec

Overhead bits = $\frac{\text{Overhead}}{\text{Frame rate}}$ = $\frac{8 \text{ kbps}}{8\text{K frames/sec}}$ = 1 bit

With overhead,

Frame Size = 24 x 8 bits + Overhead bit

= 192 bits + 1 bit

= 193 bits

Data rate = 8K frames/sec x 193 bits

= 44 Mbps

DS-2

Data rate = 96 × DS-0

= 96 × 64 kbps

= 6144 kbps

According to table,

DS-2 Data rate = 6312 kbps

Overhead = (6312 − 6144) kbps

= 168 kbps

Without Overhead,

Frame Size = 96 × 8 bits = 768 bits

Frame rate = $\frac{6144 \text{ kbps}}{768 \text{ bits}}$ = 8k frames/sec

Overhead bits = $\frac{168 \text{ kbps}}{8\text{k frames/sec}}$

= 21 bits

With overhead, Frame size = 768 + 21 = 789 bits

Data rate = 8k frames/sec × 789 bits

= 6.312 Mbps

Similarly, DS-3, DS-4

09-09-26

Ch-8 (Introduction)

Switching

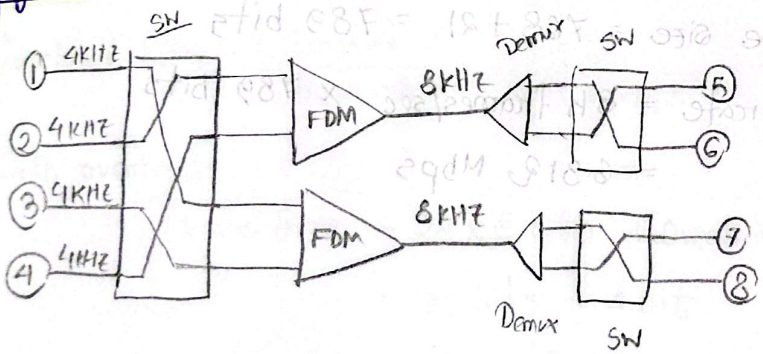
switches are interconnected.

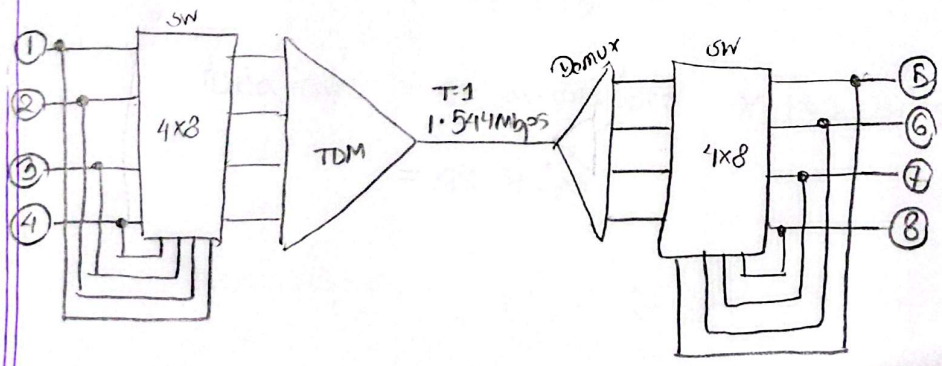
↳ are connected to the end devices

Switched network ——— message switched
× not in use

- Circuit-switched network (voice)
  - delay is not accepted
  - works in 3 phases
    - connection setup – resource allocate করা
    - data transfer – sending information through the allocated resource
    - teardown – resource free করে দেয়া
- Packet switched network (data transmission) (through internet)

eg-1



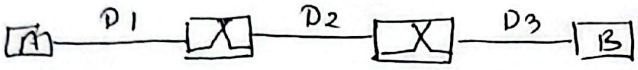


Propagation delay, $P_d = \frac{D}{S}$

Transmission delay, $T_d = \frac{L}{R}$

✓ Set up phase = Request + Acknowledgement $= (P_d + T_d) + (P_d + T_d)$

$= 2P_d + 2T_d$



Request $= \underbrace{\frac{D_1}{S} + \frac{D_2}{S} + \frac{D_3}{S}}_{P_d} + \underbrace{\frac{L}{R}}_{T_d}$

যদি A to B dist D বলে দেয়া,

তাহলে, Request $= \underbrace{\frac{D}{S}}_{P_d} + \underbrace{\frac{L}{R}}_{T_d}$

✓ Data Transfer $= P_d + T_d$

✓ Tear down $= P_d + T_d$

Problem-1

Set up & Tear down,

$L = 1000$ bits

$R = 1$ Mbps

$T_d = \frac{1000}{1 \times 10^6} = 1$ ms

$D = 5000$ km

$S = 2 \times 10^8$ m/s

$P_d = \frac{5000 \times 1000 m}{2 \times 10^8 m/s} = 25$ ms

Adv. no delay

dis adv, we get a delay in set phase before data transfer.

14-09-26

Packet switching

2 methods — datagram (independently travel করে) (sender receiver এর মধ্যে কোনো connection থাকে না)

± virtual circuit network.

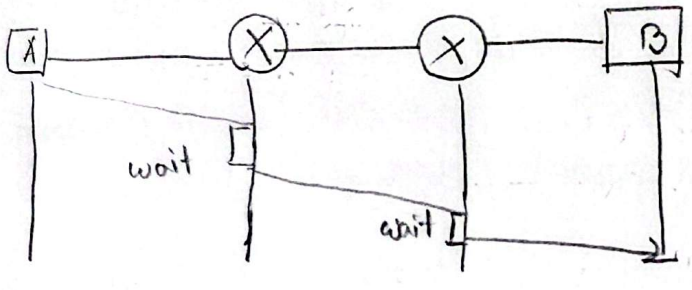
datagram—

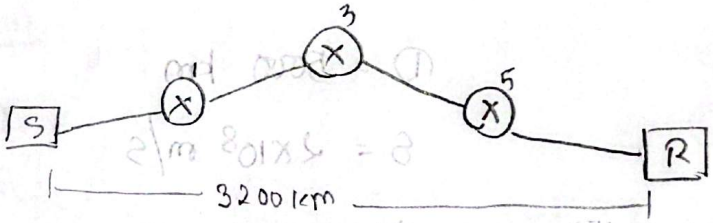
-lable থাকে না।

destination address -lable এ সকলে যে port এ যাবে

না সকলে new add করে।

Total delay = $3T_d + 3P_d + W1 + W2$





$P_d = \frac{D}{S} = \frac{3200 \times 10^3 \text{ m}}{2 \times 10^8 \text{ m/s}} =$

T_d ($\frac{1}{R}$) = ignored

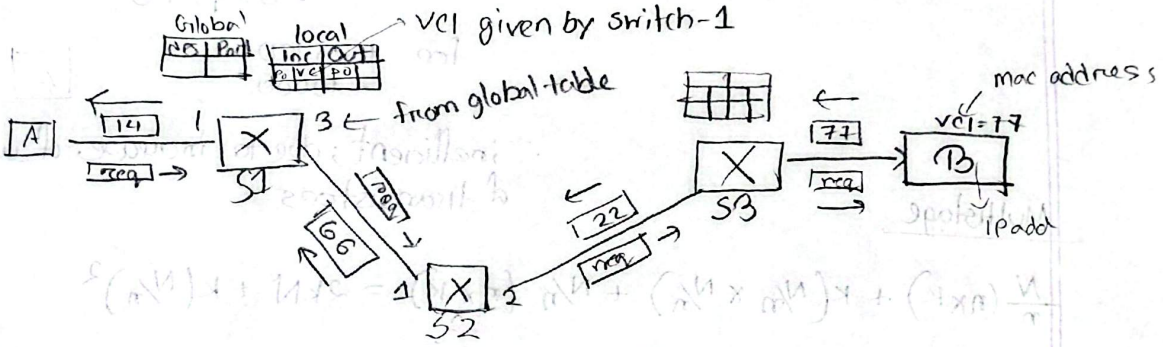
Total delay = $P_d + T_d + w_1 + w_3 + w_5$

= $D/S + 0 + 3 + 20 + 20$

Virtual circuit network —

↳ circuit - switched network + datagram network

↳ Do the allocation virtually unlike circuit switched



→ req

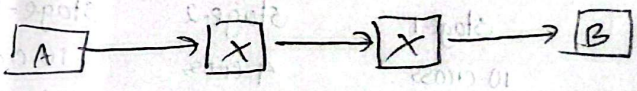
← ack

Inc		Out	
Port	VCI	Port	VCI
1	66	2	22

Assigned by switch 2

Then send
data through ports
then teardown phase

Total delay = $3T_d + 3P_d$ + setup delay + teardown delay



17-09-26

Circuit switching → space division → Cross bar
→ Multistage

Cross bar

of cross point = n x m
= 9 x 4 = 12

for 100 x 100
= 10000

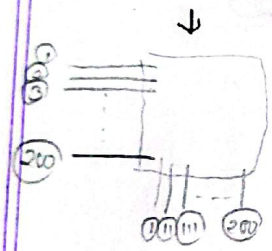
∴ inefficient; due to increase of users, increase of transistors

Multistage

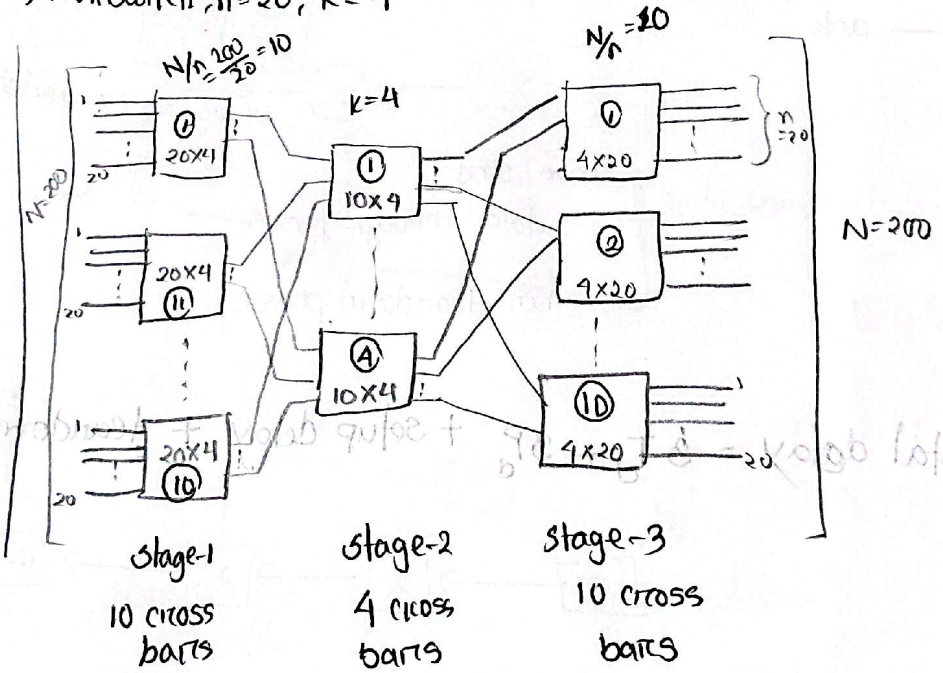
$$\frac{N}{n}(n \times k) + k\left(\frac{N}{n} \times \frac{N}{n}\right) + \frac{N}{n}(n \times k) = 2kN + k\left(\frac{N}{n}\right)^2$$

N = 200 x 200 , Multiswitch, n = 20, k = 4

cross bar



# of crosspoints
= 200 x 200
= 40000



∴ 10 (20x4) + 4 (10x10) + 10 (4x20)
= 10x80 + 4x100 + 10x80
= 800 + 400 + 800
= 2000 = 5% transistors than crossbare

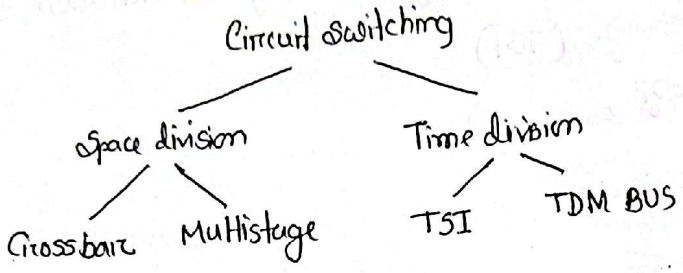
blocking, if < 25%.

This is blocking ; >25% makes non blocking. We need to bring it above 25%.

CSE 2211

17-09-26

Switch Structure



Crossbar — for increase of users, crosspoints increase ; so transistors increase; inefficient .

Multistage — 5% transistors from cross bar, but for it to become non blocking, it has to be $\geq 25\%$,

To make it non blocking ,

Clos criterion ,

$$n = \left(\frac{N}{2}\right)^{1/2}$$

$$k = 2n-1$$

Take the problem,

$n = \left(\frac{200}{2}\right)^{1/2} = 10$; $k = 2 \times 10 - 1 = 19$

Stage-1 ; $\frac{N}{n} = 20$ cross bars , 10×19 each

Stage-2 ; $k = 19$ crossbars , 20×20 each

Stage-3 ; $N/n = 20$ crossbars, 19×10 each

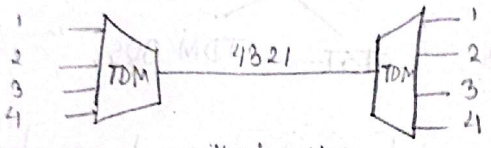
$\therefore 20(10 \times 19) + 19(20 \times 20) + 20(19 \times 10)$

$= 15200 = 25\%$

*Design the switch in multistage, following Clos criterion, for 200 X 200

Time slot interchange (TSI)

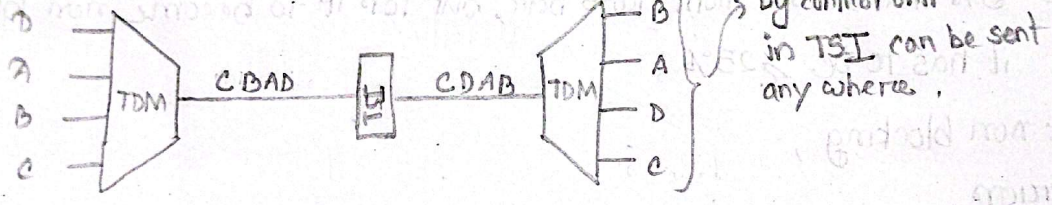
TDM ৭ use. হয়,



without switch

⇓

after switching



TSI → Control unit + RAM

↳ bits stored here.

# of Location = # of time slot

# of location size = # of bits / time slots

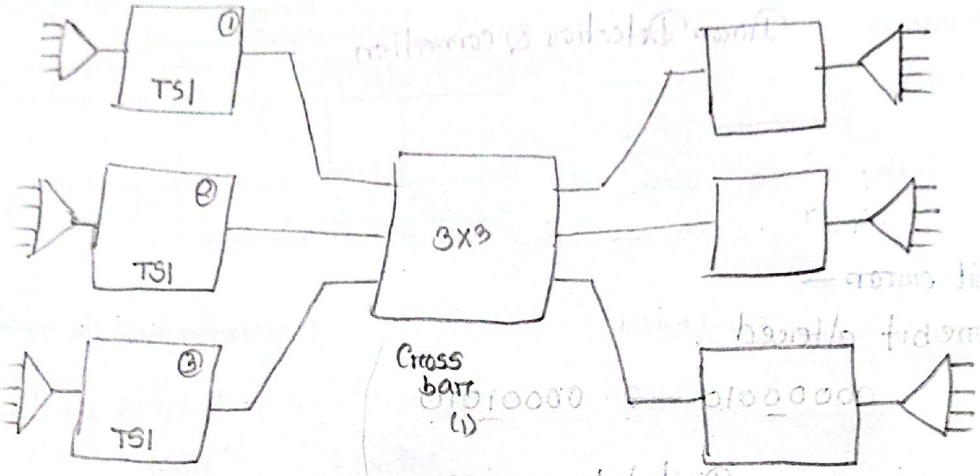
Time-Space-Time (TST)

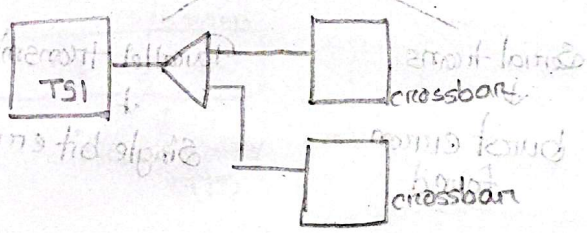
→ System non blocking

→ Crossbare decreases

for 12 X 12,

Let, n=4, k=1





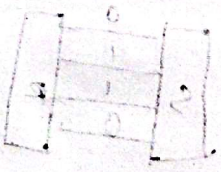
number of slots – 4

In 1st stage,
total slot number
= 3X4 = 12

In 3rd stage,
= 3X4 = 12

∴ total number of memory locations
= 12 + 12 = 24

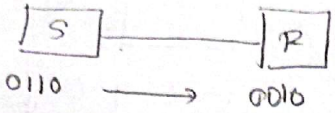
Design আকারে – Time space / Multistage



21-09-26

Ch-10

Error Detection & correction



Single bit error —

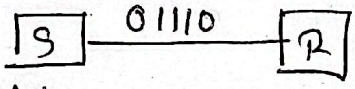
-Only one bit altered.

00000010 ⟶ 00001010

Digital transmission

- Serial trans. → burst error found
- Parallel transmission → Single bit error found

In serial transmission,

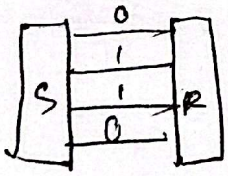


For 1 Mbps

1 bit → 1 μs

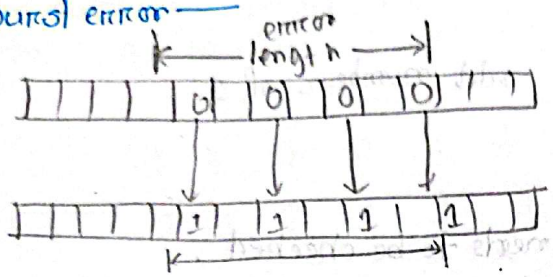
but noise stays more time, so more than one bit affected...

In parallel,



1 bit affected

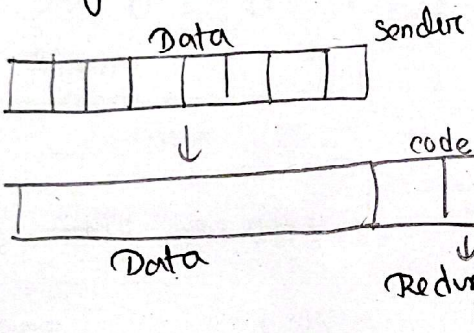
Burst error —

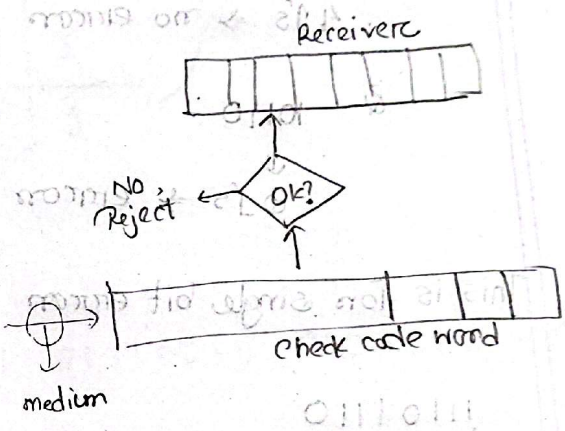


error length – from starting error to last error.

Concept of Redundancy :

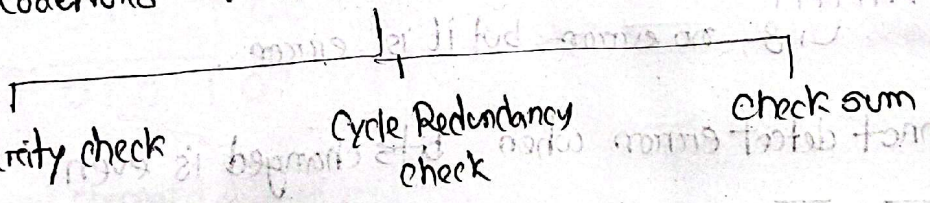
→ adding extra bit.





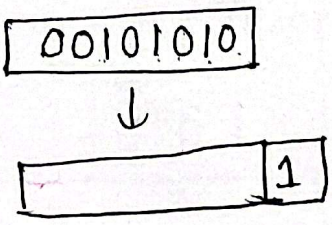
Sender, dataword ⟶ codeword

Receiver, Codeword ⟶ Error detection



Parity check :

Sender, mechanism even parity technique,



→ to make even numbers of 1.

so the one comes determined by polled.

odd parity, 01100001 → [| 0] odd number of 1

Receiver gets,

10111 and even parity needs to be checked,

↓

4 1's => no error

& 10110

↓

3 1's => error

This is for single bit error

11101110

↳ 6 so, no error

11101111

↳ 7, so error

11111111

↳ 8, no error, but it is error.

It cannot detect error when bits changed is even

odd হলে ধরবে।

01010100

↓

[1 |]

Two dimensional parity check

Sender:

	1	2	3	4				
1	1	1	0	0	1	1	1	1
2	1	0	1	1	1	0	1	1
3	0	1	1	1	0	0	1	0
4	0	1	0	1	0	0	1	1
	0	1	0	1	0	1	0	1

row AG even parity check

even parity check.

column এর even parity check

Code word: 11001111 10111011 01110010 01010011 01010101

Receiver:

1	1	0	0	1	1	1	1	✓
1	0	1	1	1	0	1	1	✓
0	1	1	1	0	0	1	0	✓ ok all no error.
0	1	0	1	0	0	1	1	✓
0	1	0	1	0	1	0	1	✓

Let, code word got changed due to error.

1	0	1	0	0	0	1	1	
1	0	0	0	1	0	0	1	×
1	1	0	1	1	1	0	1	
1	1	1	0	0	1	1	1	
1	0	1	0	1	0	1	0	
×	×	×	×	×				

error detected.

etermined by
polled.
↑
check whose inq is 1)

1 1 0 0 1
1 0 1 1
0 1 1 1 0
1

কোনো error detect হচ্ছে না।
but there is error.
-This is also limitation.

24-09-26

CRC (Cyclic Redundancy check)

- Binary format
- Polynomial format

Given → Data
→ Divisor (Common for sender & Receiver)

Binary format

Data : 1001 → Add 3-bit extra → 1001000 (n-1)
Divisor : 1011 (n)

MUL : AND
SUB : XOR

Binary Division

```
      1
1011 | 1001000
       1011
       -----
```

sender, data ⟶ codeword

Binary Division, sender

```
      1010
1011 | 1001000
       1011
       ----
        0100
        0000
        ----
         1000
         1011
         ----
          0110
          0000
          ----
           110 = R
```

∴ codeword → 1001110

Receiver)

```
1011 | 1001110
       1011
       ----
        0101
        0000
        ----
         1011
         1011
         ----
          0000
          0000
          ----
             0
```

(No error)

Remainder 0 হলে no error; otherwise error,

Let, codeword be 1101110 to receiver

```
1011 | 1101110
       1011
       ----
        1101
        1011
        ----
         1101
         1011
         ----
          1100
          1011
          ----
           111 (error)
```

determined by
polled.
(check whose irq is 1)

Polynomial format:

Data - 1001

⊛ to P ⟶ $1\times x^3 + 0\times x^2 + 0\times x^1 + 1\times x^0$

Data - $x^3 + 1$

Divisor - 1011

$1\times x^3 + 0\times x^2 + 1\times x^1 + 1\times x^0$

$= x^3 + x + 1$

↑ highest power নিয়ে data র সাথে mul.

Data - $(x^3+1)x^3 = x^6 + x^3$

Sender:

$$x^3+x+1 \;\overline{)\; x^6 + \quad + x^3 \;(\; x^3 + x}$$

$x^6 + x^4 + x^3$

x^4

$x^4 + x^2 + x$

$x^2 + x$

Receiver:

$$x^3+x+1 \;\overline{)\; x^6 + \quad + x^3 + x^2 + x \;(\; x^3 + x}$$

$x^6 + x^4 + x^3$

$x^4 + x^2 + x$

$x^4 + x^2 + x$

0

No error,

∴ codeword : $x^6 + x^3 + x^2 + x$

Ⓡ

Suppose,

Receiver got codeword $x^6 + x^4 + x^2 + x$

$$x^3+x+1 \overline{)\, x^6+x^4+x^2+x}\quad (x^3+1)$$

$x^6+x^4+x^3$

x^3+x^2+x

x^3+x+1

x^2+1 (error)

check sum

28-09-26

Sender side

Data → codeword

Data → 7, 11, 12, 0, 6 (4 bit)

7
11
12
0
6
0 ← Initially 0

sum = 36

codeword
7, 11, 12, 0, 6, 9

36 = 32 16 8 4 2 1
1 0 0 1 0 0
→ 1 0

wrapped sum = 0 1 1 0 = 6

check sum = 1001 = 9
(complement)

Receiver

codeword → Detect error
→ 7, 11, 12, 0, 6, 9

7
11
12
0
6
9

sum = 45

45 = 32 16 8 4 2 1
1 0 1 1 0 1
→ 1 0

wrapped sum = 1 1 1 1

check sum = 0000
(comp) . no error

determined by
polled.
(check whose

Receiver with errors

Code word → 7, 11, 12, 1, 6, 9

7
11
12
1
6
9
Sum = 46

46 = 1 0 1 1 1 0
→ 1 0
Wrapped sum 1 0 0 0 0
→ 1
0 0 0 1

∴ check sum = 1110 → error.

Word

Sender:

Pomoran - (16 bit) (F=70, O=111, m=114, U=117, z=122, a=97, n=110)

Fo = 4 6 6 F

mo = 7 2 6 F (Initially checksum ↓ 0000)

Uz = 7 5 7 A

an = 6 1 6 E

F = 70/16 = 16 | 70 (4), 64, 6 = (46)16

O = 111/16 = 16 | 111 (6), 96, 15(F) = (6F)16

m = 114/16 = 16 | 114 (7), 112, 2 = (72)16

1013

Fo = 4 6 F

x_0 = 7 2 6 F

ut = 7 5 7 A

an = 6 1 6 E

0 0 0 0

sum = 1 8 F C 6

1

Wrapped sum = 8 F C 7

Checksum: 7038 (15-W.S each)

Codeword =

Forouzan, 0_x(7038)

1st col

15
15
10
14
0

54 = 16 | 54 (3), 48, 6

2nd col

3
6
6
7
6
0

28 = 16 | 28 (1), 16, 12(C)

3rd col

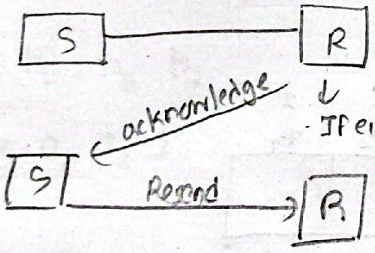
1
6
2
5
1
0

15F

4th col

4
7
7
6
0

24 = 16 | 24 (1), 16, 8



acknowledge might also get corrupted then sender gets no acknowledgement, then they dont send.

Again, while resend, error may also occur. So, this is not efficient way.

So, it is efficient for R to Detect error and correct it.

Error Correction (FEC) (Hamming Code)

Sender side

Data → code word

Data = 1001101 (k = 7)

Redundant bits = r = ? 4 = Power of 2

codeword = k + r

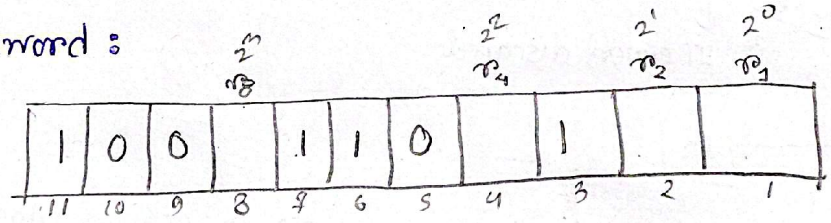
$2^r > k + r + 1$ r = 4

$16 > 7 + 4 + 1 = 12$

∴ k = 7, r = 4

code word = k + r = 11 bits

codeword :



$r_1 = b_3, b_5, b_7, b_9, b_{11}$

1 = 1, 0, 1, 0, 1

3 to 1

$r_2 = b_3, b_6, b_7, b_{10}, b_{11}$

0 = 1 1 1 0 1

$r_4 = b_3, b_5, b_7$

$0 = 0, 1, 1$

$r_8 = b_9, b_{10}, b_{11}$

$1 = 0, 0, 1$

code word

1 0 0 1 1 1 0 0 1 0 1

Receiver

1 0 0 1 1 0 0 0 1 0 1

11 10 9 8 7 6 5 4 3 2 1

$r_1 = b_3, b_5, b_7, b_{10}, b_{11}$

$1 = 1 \; 1 \; 0 \; 1 \; 0 \; 1$

$r_2 = b_3, b_6, b_7, b_{10}, b_{11}$

$0 = 1, 0, 1, 0, 1$

$r_4 = b_5, b_6, b_7$

$0 = 0, 0, 1$

$r_8 = b_9, b_{10}, b_{11}$

$1 = 0, 0, 1$

$0 \; 1 \; 1 \; 0 = 6$ ∴ 6th position is error.

we change 6.